# TALHA ALI ASHFAQ

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## PROFESSIONAL SUMMARY

Electrical Engineering graduate specializing in AI and computer vision, with hands-on experience building intelligent systems, automation solutions, and applied machine learning applications. Strong foundation in Python-based AI development, robotics, and real-world engineering problem-solving.

## EXPERIENCE

**Engineering Intern** Jul 2024 – Sep 2024
*A & N Engineering Services* *Islamabad, Pakistan*

- Gained practical experience in engineering support, technical documentation, and project coordination across live client projects.
- Collaborated with senior engineers on daily operational tasks, strengthening teamwork, communication, and analytical problem-solving skills.

## FINAL YEAR PROJECT

**AI-Augmented Food Analyzer for Visually Impaired Persons**

- Built an AI-powered mobile application that identifies food items in real time using computer vision and deep learning models.
- Integrated nutritional analysis with voice-based feedback to deliver ingredient details and calorie estimates, enabling independent, accessible decision-making for visually impaired users.

## PROJECTS

**PLC-Controlled Multi-Stage Car Wash System**

- Designed and programmed a PLC-based automation system controlling a multi-stage car wash process, including soaping, rinsing, and drying sequences.
- Implemented ladder logic with sensor-triggered stage transitions and safety interlocks to ensure reliable, sequential operation.

**Dermoscopic Image Segmentation & Feature Analysis via CCA**

- Developed an image processing pipeline to segment skin lesions from dermoscopic images using Connected Component Analysis (CCA) for region identification.
- Extracted and analyzed shape, border, and texture features from segmented lesions to support automated skin condition assessment.

**Obstacle Avoidance Robot, Line Following Robot, Master–Slave Robot**

- Built a series of embedded robotics projects applying sensor integration and control logic.

## EDUCATION

**Bahria University** Islamabad, Pakistan
*Bachelor of Engineering in Electrical and Electronics* *2021 – 2026*

**Govt. Higher Secondary School Salehphat** Sukkur, Pakistan
*Intermediate — Pre-Engineering* *2019 – 2021*

**Bahria Model School Campus 2, Majeed SRE** Karachi, Pakistan
*Matriculation — Science* *2017*

## CERTIFICATIONS

- IEEE Electromagnetic Compatibility & Interference Workshop, 2024
- Namal TECH Expo, 2026
- Zindigi Prize, 2024

## TECHNICAL SKILLS

**Programming:** Python, C/C++, Kotlin

**AI / Machine Learning:** Computer Vision, Deep Learning, Machine Learning, OpenCV, TensorFlow, Image Processing

**Engineering:** Circuit Design, Circuit Analysis, PLC Programming

**Soft Skills:** Leadership, Critical Thinking, Teamwork, Communication, Problem Solving

## LANGUAGES

Urdu (Native)
English (C2 Proficiency)